



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

ASSIGNMENT: 4

Agile devops and development lab

NAME : SHARU SREE.S

REG.NO : 23MIS0403

SLOT : L3+L4

FACULTY : SWETA BHATACHARYA

Create an HTML file for your profile and complete the following:

1. Create a repository in GIT and create the following files:

- **index.html**
- **Dockerfile • Deployment.yaml**

index.html:

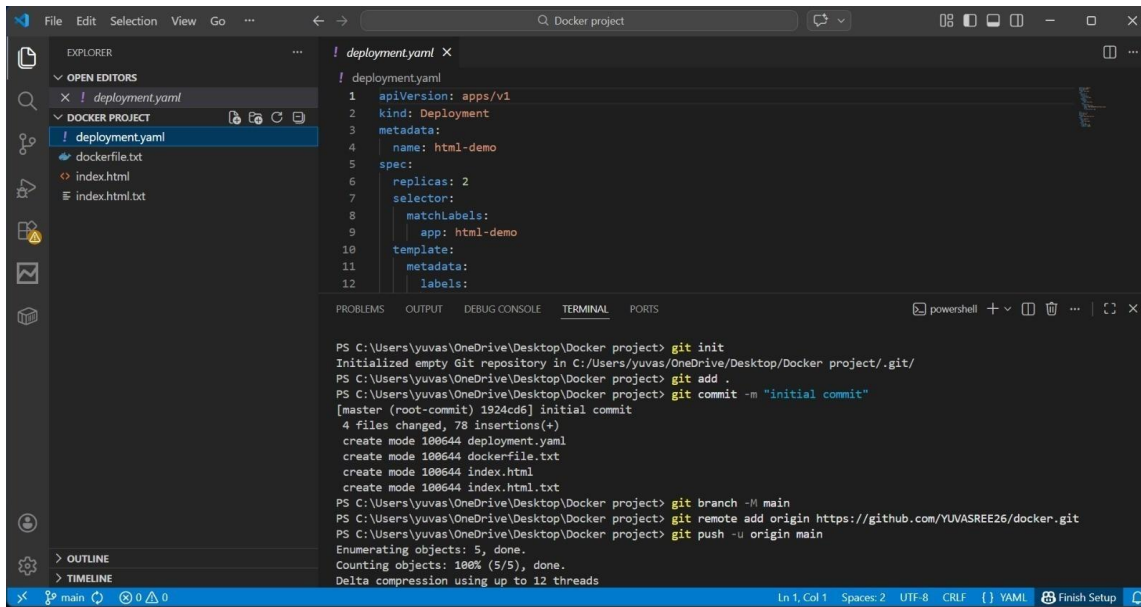
```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center; font-
family:Times New Roman; background-color:#af0d0d;
} h1 {
color:#0ef
777; }
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body> </html> dockerfile:
```

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
```

EXPOSE 80 deployment.yaml:

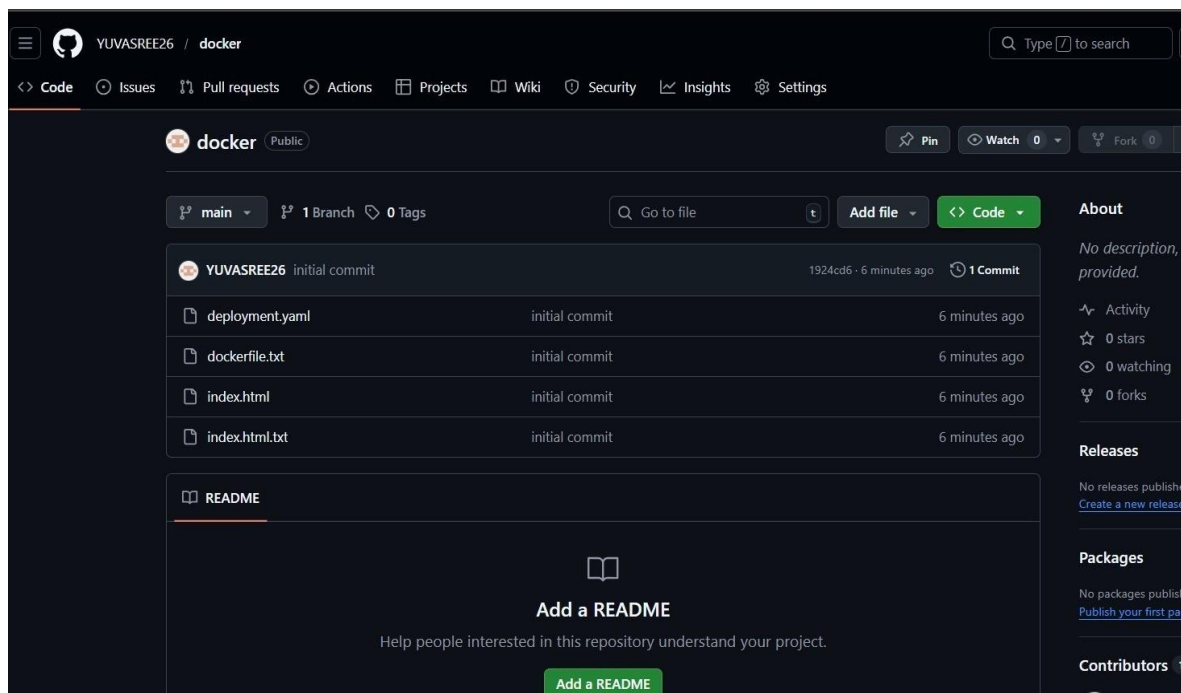
```
apiVersion: apps/v1 kind:
Deployment metadata:
name: html-demo spec:
  replicas: 2 selector:
matchLabels:
app: html-demo
template:
metadata: labels:
app: html-demo
spec: containers:
- name: html-demo image: yuva26/html-demo:latest ports:
- containerPort: 80
```

```
--apiVersion: v1 kind: Service
metadata: name: html-
demo-service spec: type:
NodePort selector:
  app: html-demo ports:
- port: 80 targetPort: 80
nodePort: 30010
```

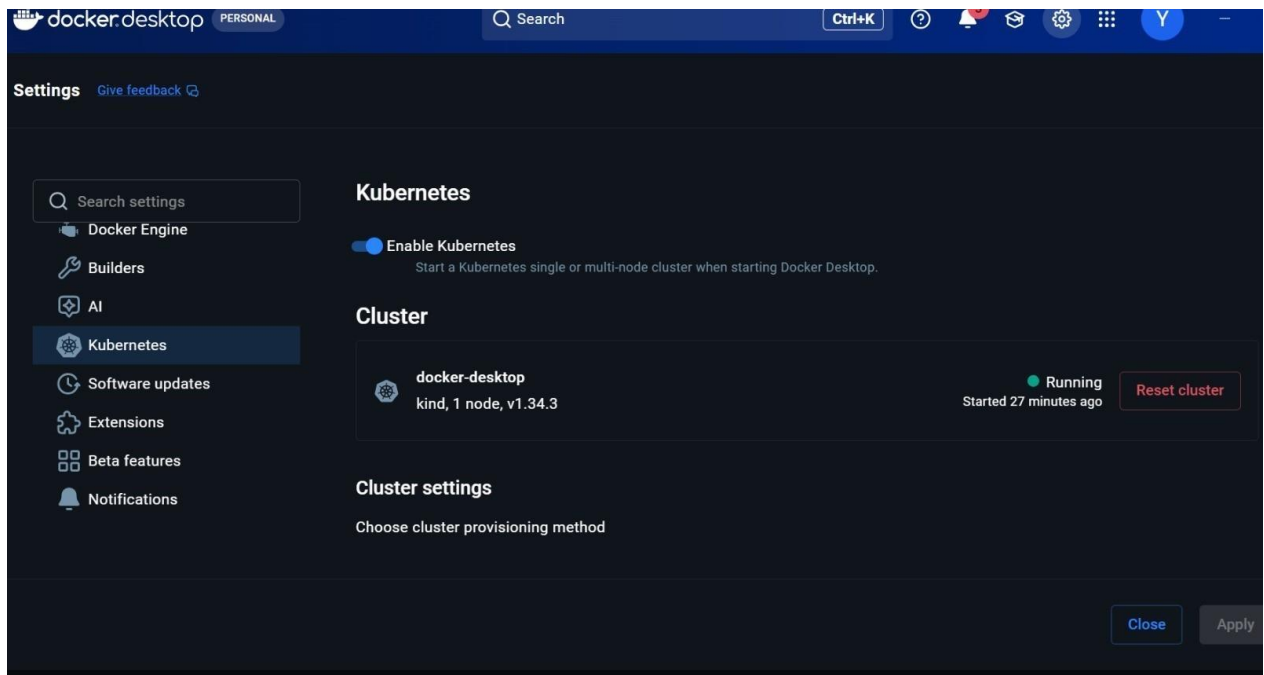


```
! deployment.yaml
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: html-demo
5 spec:
6   replicas: 2
7   selector:
8     matchLabels:
9       app: html-demo
10  template:
11    metadata:
12      labels:
```

```
PS C:\Users\yuv\OneDrive\Desktop\Docker project> git init
Initialized empty Git repository in C:\Users\yuv\OneDrive\Desktop\Docker project\.git\
PS C:\Users\yuv\OneDrive\Desktop\Docker project> git add .
PS C:\Users\yuv\OneDrive\Desktop\Docker project> git commit -m "initial commit"
[master (root-commit) 1924cd6] initial commit
 4 files changed, 78 insertions(+)
 create mode 100644 deployment.yaml
 create mode 100644 dockerfile.txt
 create mode 100644 index.html
 create mode 100644 index.html.txt
PS C:\Users\yuv\OneDrive\Desktop\Docker project> git branch -M main
PS C:\Users\yuv\OneDrive\Desktop\Docker project> git remote add origin https://github.com/YUVASREE26/docker.git
PS C:\Users\yuv\OneDrive\Desktop\Docker project> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
```

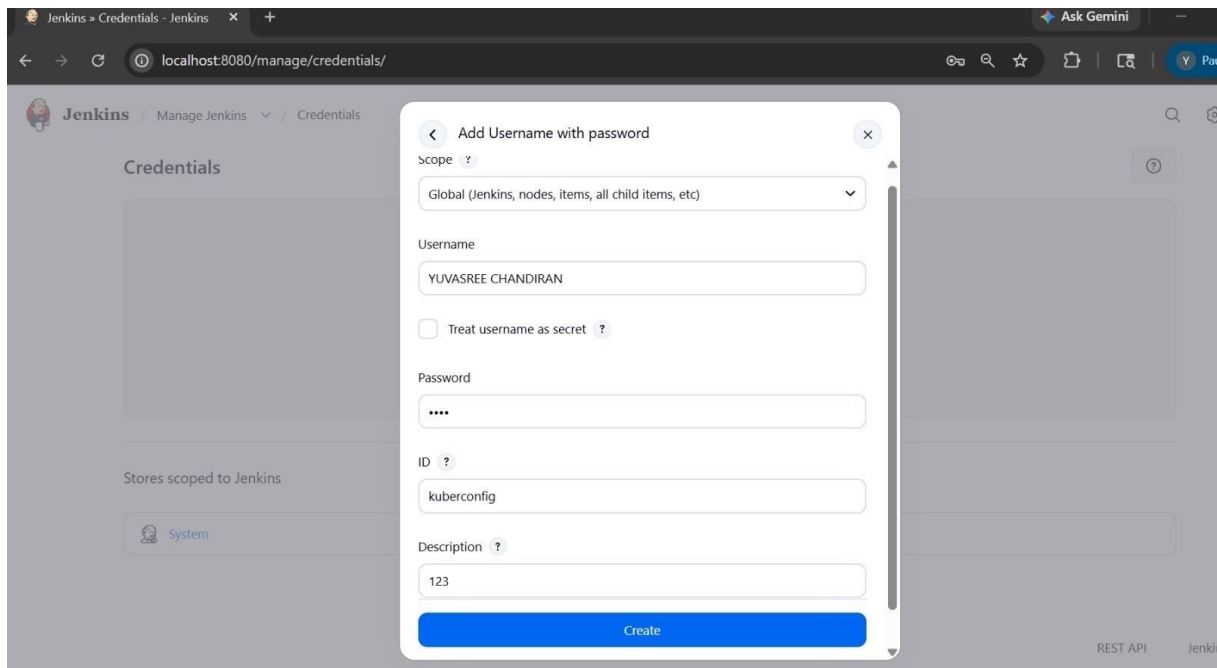


2.Enable Kubernetes



3.Open Jenkins

In Jenkins Setting --> Credentials



System » Global - Jenkins

localhost:8080/manage/credentials/store/system/domain/_/

Jenkins / Manage Jenkins / Credentials / System / Global

Global

Credentials that should be available everywhere.

+ Add Credentials

YUVASREE CHANDIRAN/***** (123)

kuberconfig - 123

Choose file No file chosen

ID ?

kubeconfig

Description ?

kubernetes credential

Create

```
pipeline {
  agent any
  environment {
    DOCKER_IMAGE = "yuva26/html-demo"
  }
  stages {
    stage('Clone Code') {
      steps {
        git branch: 'main', url: 'https://github.com/YUVASREE26/docker.git'
      }
    }
    stage('Build Docker Image') {
      steps {
        bat 'docker build -t %DOCKER_IMAGE% .'
      }
    }
  }
}
```

```

} } stage('Push Image') { steps
{ withCredentials([usernamePassword(credentialsId:
'dockerhub', usernameVariable: 'USER', passwordVariable:
'PASS')) { bat 'docker login -u %USER% -p %PASS%' bat
'docker push %DOCKER_IMAGE%'
}
} } stage('Deploy to Kubernetes') { steps {
withCredentials([file(credentialsId: 'kuberconfig',
variable:
'KUBECONFIG')])) {
bat ""
set KUBECONFIG=%KUBECONFIG%
kubectl apply -f deployment.yaml --validate=false
""
}
}
}
}
}
}
}
}

```

✓ cicd demo

Stage View

	Clone Code	Build Docker Image	Push Image	Deploy to Kubernetes
Average stage times: (full run time: ~22s)	1s	3s	4s	169ms
#3 Mar 26 15:26 1 commit	1s	8s	12s	392ms

